

RIFAMPICIN-RESISTANT TB (RR-TB): ROUTINE CARE

- RR-TB refers to TB that is resistant to rifampicin, with or without resistance to other TB medications. If patient has INH mono-resistant TB →95.
- **If RR-TB care not available, refer to closest drug-resistant TB unit.**

- **If newly diagnosed with RR-TB or pre-XDR TB:** if pretomanid available, consult the updated 'Clinical Management of RR-TB, September 2023' guideline to start patient on new short 6-mth BPaLL (or BPaL if pre-XDR) regimen.
- **Note:** 1st and 2nd LPA tests are being replaced with Xpert MTB/XDR assays for rapid detection of resistance to isoniazid, fluoroquinolones, amikacin and ethionamide.

Assess the patient with RR-TB

Assess	When to assess	Note
Registration	Every visit	Enter patient's details at diagnosis. Update register (EDR.web) with latest sputum results at every visit.
Symptoms	Every visit	• If respiratory rate ≥ 30, breathless at rest or while talking, prominent use of breathing muscles, drowsy/confused, coughs up ≥ 1 tablespoon fresh blood, neck stiffness, persistent vomiting or new weakness of arm/leg, give urgent attention →92. If persistent episodes of coughing blood, consider referral for surgical review. • Expect gradual improvement. If not improving, assess adherence → 173 and review LPA and DST results. • If still no improvement at 4 months, request 1st and 2nd line LPA and extended phenotypic DST and present to NCAC¹ to advise on rescue regimen.
Adherence	Every visit	Check patient is attending clinic daily for treatment (or on appointment day if receiving supply of medications).
Side effects	Every visit	• Ask about side effects of treatment → 105. Manage promptly as side effects are common cause of treatment interruption. • If intolerance to any medication, present to PCAC²/NCAC for medication substitution. Email or fax adverse drug reaction (ADR) form to adr@sahpra.org.za or (012) 842 7609/10.
TB contacts	At diagnosis	• Ask if patient is a TB contact³ of index patient with RR-TB. If yes, check contact's LPA and DST results to help decide patient's RR-TB treatment regimen. • Advise that all TB contacts³ visit the clinic for TB screening/prevention.
Family planning	Every visit	• Advise to avoid pregnancy during treatment, assess patient's contraceptive needs → 154. If on injectable contraceptive, no need to change interval between doses. • If pregnant → 159 and present to NCAC. Avoid delaying treatment, start while awaiting response.
Depression	Every visit	In the past month, has patient: 1) felt down, depressed, hopeless or 2) felt little interest or pleasure in doing things? If yes to either → 143.
Alcohol/drug use	At diagnosis, 4 months	In the past year, has patient: 1) drunk ≥ 4 drinks ⁴ /session, 2) used illegal drugs or 3) misused prescription or over-the-counter medications? If yes to any → 142.
Palliative care	If deteriorating	If patient breathless at rest, unable to walk unaided or failing treatment, also consider giving palliative care → 170.
Weight (BMI ⁵)	Every visit	Expect weight gain on treatment and adjust treatment doses. If losing weight on treatment, discuss with specialist/refer. If BMI < 18.5 , refer for nutritional support.
BP	At diagnosis	If known hypertension → 133. If not, check BP: if $\geq 140/90$ → 132.

Check routine tests according to table and review results →100:

At diagnosis	At 2 weeks	At 4 weeks and then monthly	At 3 months	At 6 months	At 12 months	Other
• 1 sputum for DR-TB reflex DST testing (smear, culture, 1st and 2nd line LPA, phenotypic DST) • ECG, chest x-ray • Vision (Snellen chart) • Pregnancy test • HIV → 110 • Fingerprick glucose • FBC, differential count, ALT, creatinine, potassium, magnesium, TSH • If HIV: CD4, viral load	• If on linezolid: FBC, differential count	• If pulmonary TB: 1 sputum for TB microscopy and culture • If on bedaquiline, clofazimine, moxifloxacin or delamanid: ECG • If on linezolid: FBC, differential count, vision (Snellen chart) • If on amikacin: audiometry, creatinine, potassium, magnesium	• HIV → 110 • If on ethionamide or PAS: TSH	• Chest x-ray • If HIV: CD4, viral load	• HIV → 110 • If HIV: CD4, viral load	• If on amikacin: baseline audiometry (hearing test) • Once bedaquiline stopped: ECG 3 monthly • If HIV: viral load 6 monthly • If on ethionamide or PAS: TSH 3 monthly • If unwell: chest x-ray, ALT, Creat, K⁺, Mg

Review results →100.

¹National Clinical Advisory Committee: NCAC@witshealth.co.za. ²Provincial Clinical Advisory Committee. ³A TB contact refers to a patient who shared an enclosed space (at work, socially, in a hostel, or in a household setting), for ≥ 1 night or for frequent/extended daytime periods, with an adult/adolescent with pulmonary TB ("index patient"), during the 3-month period before the index patient started their TB treatment. ⁴One drink is 1 tot of spirits, or 1 small glass (125mL) of wine or 1 can/bottle (330mL) of beer. ⁵BMI = weight (kg) ÷ height (m) ÷ height (m).